GPU

GPU

1.1

- D–T 14 MeV
-
- [1–2]

1.2

- CAD/CSG CAD [1]
-
- GPU

1.3 MOC GPU

- MOC OpenMOC GPU-MOC MOC [3–6]
- GPU
-

2.1

1. CSG CAD
2. OpenMOC MOCkingbird MOC 2D/1D [3,6]
3. DDA [9]
4.

2.2 MOC GPU

1. MOC GPU GPU [4–5,8] OpenMOC [3]

2. – [6,8]

3. NuDEAL MOC/HFEM DGMOC DFEM-SN
GPU-MOC[7]

4. Roofline [10]

2.3

- Baker [11–12]
-
- GPU

2.4

- CADIS FW-CADIS [13–14]
- FW-CADIS [15]
- FW/CADIS-Ω [16]
- Raffuzzi [17]
  OpenMC FW-CADIS [18]

2.5

- VITAS – [19–20]
- Yin heterogeneous GPU [21]
- Sun UVNM-SN HDG [22–23]
- Xiao Zhang VITAS [24]
  GPU

3.1

GPU

3.2

-
-
-

3.3

-
-

-

3.4

-
- GPU-MOC
- FW-CADIS

3.5

GPU-MOC – –

1% 3% 5%

4.1

CAD/CSG → → → MOC → GPU → →

4.2

1.

2. / /

3.

4. /

5.

6.

4.3 MOC GPU

(n,xn) FENDL

[2]

–

– –

CPU GPU GPU

GPU

4.4

1. / —
2. CADIS FW-CADIS [13–14]
3.
4. MC /MAGIC FW-CADIS [17–18]
5. FOM = 1/(r²T_total) r T_total / FOM
6. — FW/CADIS-Ω [16]

4.5

层次	算例与比较	回答的问题
	MC MC	
	MOC MC	
	FNG/FNS	
	CPU/GPU /	

(n,xn)

MOC C5G7

5.1

5.2 MOC

GPU DDA

GPU

5.3

6.1

MOC GPU [3–18] CPU/GPU
MOCkingbird NuDEAL VITAS

6.2

课题组已发表工作	可衔接的方法问题	本课题拟推进的部分
VITAS [19–20]		
VNM [21]		GPU
UVNM-SN [22]		
HDG[23]		— MOC
HVNM [24]		

GPU

6.3

-
- /
- **GPU** GPU
-
- GPU

阶段	主要任务	阶段成果与推进条件
1	CPU MOC	
2	GPU	
3		GPU
4	CADIS/FW-CADIS	

GPU-MOC —

2026 10 18

GB/T 7714

[1] Valentine A, Berry T, Bradnam S, et al. Benchmarking of emergent radiation transport codes for fusion neutronics applications. Fusion Engineering and Design, 2022, 180: 113197. https://doi.org/10.1016/j.fusengdes.2022.113197

[2] Schnabel G, Aldama D L, Bohm T, et al. FENDL: A library for fusion research and applications. Nuclear Data Sheets, 2024, 193: 1–78. https://doi.org/10.1016/j.nds.2024.01.001

[3] Boyd W, Shaner S, Li L, Forget B, Smith K. The OpenMOC method of characteristics neutral particle transport code. Annals of Nuclear Energy, 2014, 68: 43–52. https://doi.org/10.1016/j.anucene.2013.12.012

[4] Zhang Z, Wang K, Li Q. Accelerating a three-dimensional MOC calculation using GPU with CUDA and two-level GCMFD method. Annals of Nuclear Energy, 2013, 62: 445–451. https://doi.org/10.1016/j.anucene.2013.06.039

[5] Song P, Zhang Z, Liang L, Zhang Q, Zhao Q. Implementation and performance analysis of the massively parallel method of characteristics based on GPU. Annals of Nuclear Energy, 2019, 131: 257–272. https://doi.org/10.1016/j.anucene.2019.02.026

[6] Gaston D R, Forget B, Smith K S, et al. Method of Characteristics for 3D, Full-Core Neutron Transport on Unstructured Mesh. Nuclear Technology, 2021, 207(7): 931–953. https://doi.org/10.1080/00295450.2021.1871995

[7] Kim K M, Im J, Lee H G, Jung Y S. Verification and Performance Assessment of NuDEAL, a GPU-Accelerated Deterministic Transport Framework on Unstructured Meshes. Nuclear Science and Engineering, 2026, online publication. https://doi.org/10.1080/00295639.2026.2652773 . Author manuscript: https://arxiv.org/abs/2607.01591

[8] Zhou F, Li S, Xue R, et al. Neutron particle transport 3D method of characteristic Multi GPU platform Parallel Computing. arXiv preprint, 2025, arXiv:2503.17743. https://arxiv.org/abs/2503.17743

[9] Amanatides J, Woo A. A Fast Voxel Traversal Algorithm for Ray Tracing. Eurographics 1987: 3–10. https://doi.org/10.2312/egtp.19871000

[10] Williams S, Waterman A, Patterson D. Roofline: An insightful visual performance model for multicore architectures. Communications of the ACM, 2009, 52(4): 65–76. https://doi.org/10.1145/1498765.1498785

[11] Baker C M J, Buchan A G, Pain C C, et al. Goal based mesh adaptivity for fixed source radiation transport calculations. Annals of Nuclear Energy, 2013, 55: 169–183. https://doi.org/10.1016/j.anucene.2012.11.029

[12] Wang X, Zhang B, Chen Y, Xiong J. Verification of the Discrete Ordinates Goal-Oriented Multi-Collision Source Algorithm with Neutron Streaming Problems. Energies, 2022, 15(22): 8335. https://doi.org/10.3390/en15228335

[13] Wagner J C, Haghighat A. Automated Variance Reduction of Monte Carlo Shielding Calculations Using the Discrete Ordinates Adjoint Function. Nuclear Science and Engineering, 1998, 128(2): 186–208. https://doi.org/10.13182/NSE98-2

[14] Wagner J C, Peplow D E, Mosher S W. FW-CADIS Method for Global and Regional Variance Reduction of Monte Carlo Radiation Transport Calculations. Nuclear Science and Engineering, 2014, 176(1): 37–57. https://doi.org/10.13182/NSE12-33

[15] Ibrahim A M, Wilson P P, Sawan M E, et al. Assessment of fusion facility dose rate map using mesh adaptivity enhancements of hybrid Monte Carlo/deterministic techniques. Fusion Engineering and Design, 2014, 89(9–10): 1875–1879. https://doi.org/10.1016/j.fusengdes.2014.02.046

[16] Munk M, Slaybaugh R N, Pandya T M, Johnson S R, Evans T M. FW/CADIS-Ω: An angle-informed hybrid method for deep-penetration radiation transport. arXiv preprint, 2016, arXiv:1612.00793. https://arxiv.org/abs/1612.00793

[17] Raffuzzi V, Neame R, Cosgrove P. Application of the Random Ray Method to Variance Reduction in Radiation Shielding Problems. Nuclear Science and Engineering, 2026, 200(sup1): S485–S502; first published online 2025. https://doi.org/10.1080/00295639.2025.2458958

[18] OpenMC Development Team. Variance Reduction; Random Ray. OpenMC Documentation, stable version, accessed 2026-10-18. https://docs.openmc.org/en/stable/methods/variance_reduction.html ; https://docs.openmc.org/en/stable/methods/random_ray.html

[19] Zhang T, Xiao W, Yin H, Sun Q, Liu X. VITAS: A multi-purpose simulation code for the solution of neutron transport problems based on variational nodal methods. Annals of Nuclear Energy, 2022, 178: 109335. https://doi.org/10.1016/j.anucene.2022.109335

[20] , , , . VITAS . , 2023, 44(2): 15–23. https://doi.org/10.13832/j.jnpe.2023.02.0015

[21] Yin H, Liu X J, Zhang T F. An efficient parallel algorithm of variational nodal method for heterogeneous neutron transport problems. Nuclear Science and Techniques, 2024, 35: 69. https://doi.org/10.1007/s41365-024-01430-4

[22] Sun Q, Liu X, Chai X, He H, Zhang T. A discrete-ordinates variational nodal method for heterogeneous neutron Boltzmann transport problems. Computers & Mathematics with Applications, 2024, 170: 142–160. https://doi.org/10.1016/j.camwa.2024.06.032

[23] Sun Q, Liu X, Chai X, et al. Hybrid discontinuous Galerkin method for the hyperbolic linear Boltzmann transport equation for multiscale problems. Physical Review E, 2024, 110: 065301. https://doi.org/10.1103/PhysRevE.110.065301

[24] Xiao W, Zhang T. On the treatment of anisotropic scattering in the heterogeneous variational nodal method. Transactions of the Korean Nuclear Society Autumn Meeting, Gyeongju, Korea, October 26–27, 2023. https://www.kns.org/files/int_paper/paper/RPHA-2023_2023_5/RPHA-2023-WeiXiao.pdf